EE3042/3014 Midterm Exam **Date: 04 May, 2023**

Student Name and Number _____ Signature _____

The notes and books will be kept closed. The calculator can be used. The required formulas are given.

1. (25p) Three-phase 20-kVA 24,000/277-V distribution transformer is connected in $\Delta - Y$. The open-circuit test was performed on the low-voltage side of this transformer bank, and the following data were recorded: $V_{line,OC} = 480\ V$, $I_{line,OC} = 4.05\ A$, $P_{3\emptyset,OC} = 948\ W$.
   The short-circuit test was performed on the high-voltage side of this transformer bank, and the following data were recorded: $V_{line,SC} = 1400\ V$, $I_{line,SC} = 1.80\ A$, $P_{3\emptyset,SC} = 915\ W$
   (a) Find the per-unit equivalent circuit of this transformer bank.
   (b) Find the voltage regulation of this transformer bank at the rated load and 0.88 PF lagging.
   (c) What is the transformer bank's efficiency under these conditions?
2. (25p) A linear machine shown in figure has magnetic flux density of 0.5 T directed into the page, a resistance of 0.25Ohms, a bar length of l=1.0m, and a battery of 100 V.
   (a) What is the initial force on the bar at starting? What is the initial current flow?
   (b)What is the no-load steady-state speed of the bar?
   (c) If the bar is loaded with a force of 25 N opposite to the direction of motion, what is the new steady-state speed? What is the efficiency of the machine under these circumstances?
   (d) Draw the plots for the velocity, induced force, induced voltage, and current as a function of time in the machine for parts (b), (c) and (d).
3. (25p) A 100-hp, 250-V, 1200 r/min shunt de motor with an armature resistance of 0.03 Ohms and a field resistance of 41.67 Ohms. The motor has compensating windings, so armature reaction can be ignored. Mechanical and core losses may be assumed to be negligible for the purposes of this problem. The motor is assumed to be driving a load with a line current of 126 A and an initial speed of 1103 r/min. To simplify the problem, assume that the amount of armature current drawn by the motor remains constant. If the machine's magnetization curve is shown in Figure 1, what is the motor's speed if the field resistance is raised to 50 Ohms?
4. (25p) A separately excited de generator is rated at 172 kW. 430 V, 400 A, and 1800 r/min. Its magnetization curve is shown in Figure 2. This machine has the following characteristics:

$R_A = 0.05\ \Omega$ $V_F = 430$ V
$R_F = 20\ \Omega$ $N_F = 1000$ turns per pole
$R_{adj} = 0$ to $300\ \Omega$

   (a)If the variable resistor Radj in this generator's field circuit is adjusted to 63 Ohms and the generator's prime mover is driving it at 1600 r/min, what is this generator's no-load terminal voltage?
   (b)What would its voltage be if a 360-A load were connected to its terminals? As- sume that the generator has compensating windings.
   (c) What would its voltage be if a 360-A load were connected to its terminals but the generator does not have compensating windings? Assume that its armature reaction at this load is 450 A • turns.
   (d) How much field current would be needed to restore the terminal voltage to its no-load value? (Assume that the machine has compensating windings.) What is the required value for the resistor Radj to accomplish this?

Table 1. Formulas

$P = E_A I_A$	$Q = V_s I_s sin\theta$ per phase	$P = \sqrt{3} V_s I_s cos\theta$	$F_{net} = ma$
$F = i(lxB)$	$VR = \frac{V_{P,pu} - V_{S,fl,pu}}{V_{S,fl,pu}}$	e_{ind} =(vXB).l	$\eta = \frac{P_{out}}{P_{in}} x100$
$E_A = K\emptyset w$	$Pout = w\,\tau$	$S_{base} = V_{base} I_{base}$	$Y_{EX} = G_C - jB_M$

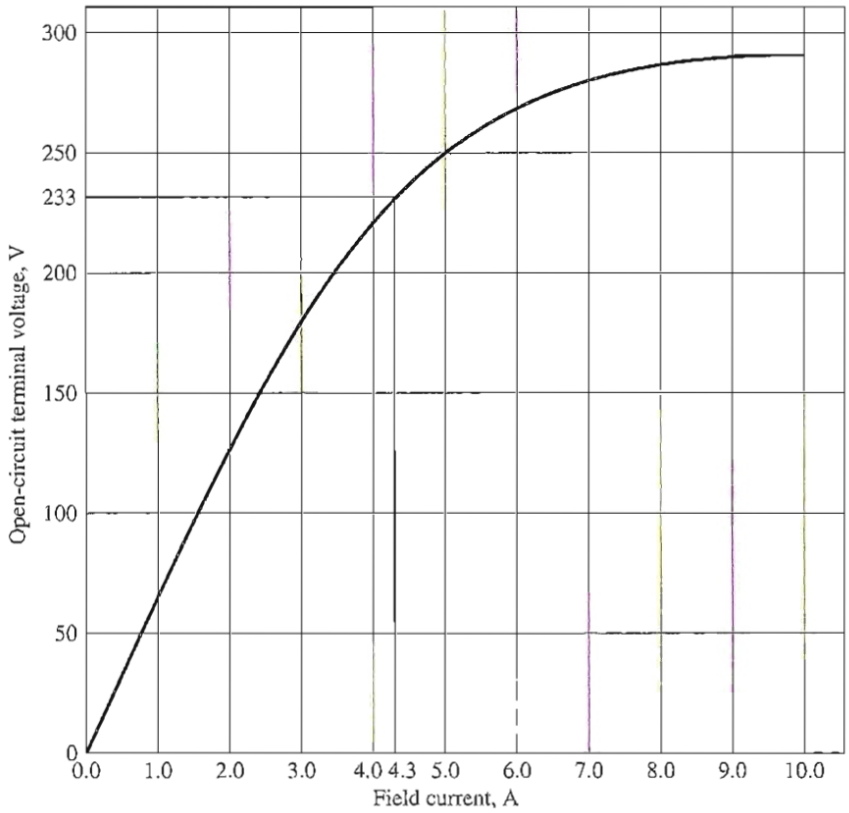


Figure 1. The magnetization curve of a typical 250-V dc motor, taken at a speed of 1200 r/min (Q3).

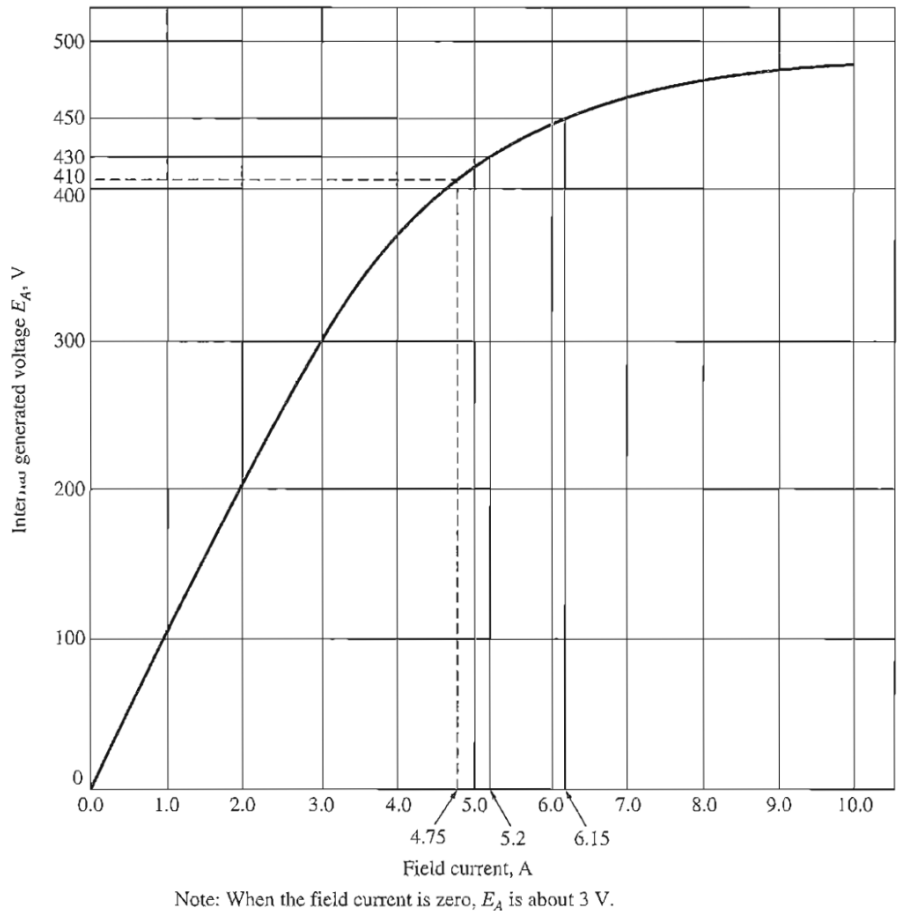


Figure 2. Magnetization curve for the generator (Q4).

ANSWER SHEET 2023 SPRING

1) 20-kVA, 24,000/277-V DT, Δ-Y

LVS ⇒ $V_{line,OC} = 480V$, $I_{line,OC} = 4.05A$, $P_{3\phi,OC} = 948W$, $I_{line,SC} = 1.80A$, $P_{3\phi,SC} = 915W$, $V_{line,SC} = 1400V$, HVS

a) per unit equivalent circuit

b) VR at rated load and 0.9 PF lagging

c) η% ?

a) $|Y_{EX}| = \frac{I_{\phi,OC}}{V_{\phi,OC}} = \frac{4.05A}{277V} = 0.0146\ S$

The angle $\theta = -\cos^{-1}\left(\frac{P_{\phi,OC}}{V_{\phi,OC} I_{\phi,OC}}\right) = -\cos^{-1}\left(\frac{948/3}{(277)(4.05)}\right)$

$\Rightarrow \theta = -73.64°$

LVS ⇒ $Y_{EX} = G_C - jB_M = 0.0146\angle -73.64° = 0.004112 - j0.014$

$\Rightarrow R_C = 243.2\ \Omega$, $X_M = \frac{1}{B_M} = 71.423\ \Omega$, $V_{\phi,base} = \frac{277}{\sqrt{3}}$

$Z_{base} = \frac{(V_{\phi,base,S})^2}{S_{\phi,base}/3} = \frac{(277/\sqrt{3})^2}{20kVA/3} = 3.836\ \Omega$

$R_C = \frac{243.2}{3.836} = 63.4\ pu$. $X_M = \frac{71.423}{3.836} = 18.6\ pu$

HVS ⇒ $V_{\phi,SC} = V_{L,SC} = 1400V$, $I_{\phi,SC} = I_{L,SC}/\sqrt{3} = 1.04A$, $P_{\phi,SC} = 915/3 = 305W$

$Z_{EQ} = \frac{V_{\phi,SC}}{I_{\phi,SC}} = \frac{1400V}{1.04A} = 1346.2\ \Omega$

$\theta = \cos^{-1}\left(\frac{P_{\phi,SC}}{V_{\phi,SC} I_{\phi,SC}}\right) = \cos^{-1}\left(\frac{305W}{(1400)(1.04)}\right) = 77.91°$

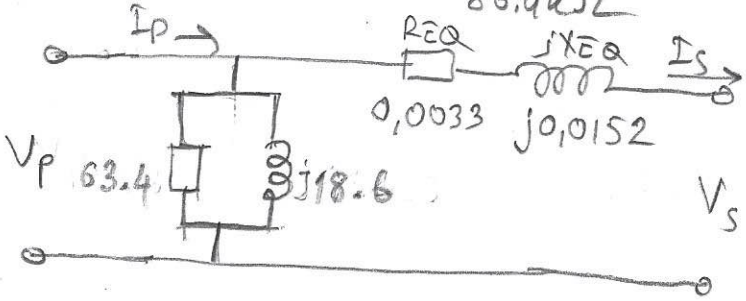
$Z_{EQ,P} = R_{EQ,P} + jX_{EQ,P} = 1346.2\angle 77.91 = 281.96 + j1316.34\ \Omega$

HVS

$Z_{base,P} = \frac{(V_{\phi base})^2}{S_\phi/3} = \frac{(24000)^2}{20kVA/3} = 86,400\ \Omega = 86.4k\Omega$

$$R_{EQ,P} = R_{EQ} = \frac{281{,}96\,\Omega}{86.4\,k\Omega} = 0{,}0033\ pu$$

$$X_{EQ,P} = X_{EQ} = \frac{1316{,}34\,\Omega}{86.4\,k\Omega} = 0{,}0152\ pu$$



b) 0,90 PF lagging $\Rightarrow -\cos^{-1}(0{,}88) = -28{,}36°$

$$V_P = V_S + I_S Z_{EQ} = 1{,}0\angle 0° + (1\angle -28{,}36°)(0{,}0033 + j\,0{,}0152)$$

$$V_P = 1{,}0\angle 0° + (1\angle -28{,}36)(0{,}00363\angle 24{,}73) = 1{,}0036\angle -0{,}01312\ pu$$

$$VR = \frac{V_{P,pu} - V_{S,pu}}{V_{S,pu}} \times 100 = \frac{1{,}0036 - 1.0}{1.0} \times 100 = 0{,}36\,\%$$

c) $P_{out_{pu}} = V_S I_S \cos\theta = (1.0)^2(1.0)(0{,}88) = 0{,}88\ pu$

$$P_{cu_{pu}} = I_S^2 R_{EQ} = (1.0)^2(0{,}0033) = 0{,}0033\ pu$$

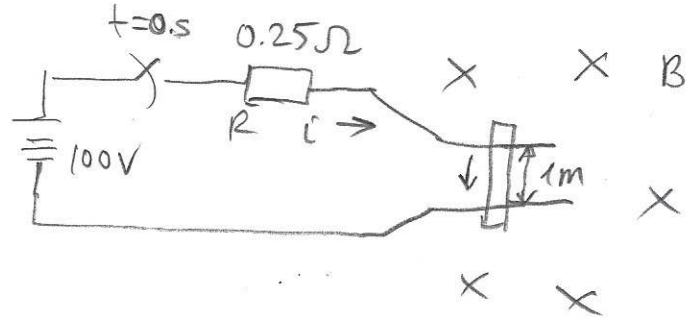
$$P_{core_{pu}} = \frac{V_P^2}{R_C} = \frac{(1.0036)^2}{53.4} = 0{,}01588$$

$$P_{in_{pu}} = P_{out} + P_{cu} + P_{core} = 0{,}88 + 0{,}0033 + 0{,}01588\ pu$$

$\Rightarrow P_{in} = 0{,}899$

$$\eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{0{,}88}{0{,}899} \times 100 = 0{,}9786\,\%$$

2)



a) $F = ?$ $\quad i = \frac{V_B}{R} = \frac{100V}{0,25\Omega} = 400$, At starting

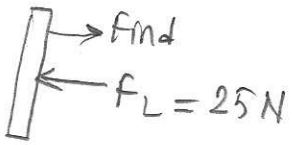
$F = i(\ell \times B) = i\ell B = (400)(1m)(0,5T) = 200N$, to the right (LHR)

b) At steady-state $\varepsilon_{ind} = V_B = v_{ss} B\ell \Rightarrow V_{ss} = \frac{V_B}{B\ell} = \frac{100V}{(0,5T)(1m)}$

$v_{ss} = ?$

$\Rightarrow V_{ss} = 200\ m/s$

c) $F_L = 25N$



$F_L = F_{ind} = i\ell B = 25N \Rightarrow i = \frac{25N}{(1m)(0,5T)} = 50A$

$\varepsilon_{ind} = V_B - iR = 100V - (50A)(0,25\Omega) = 87,5\ V$

$V_{ss,new} = \frac{\varepsilon_{ind}}{B\ell} = \frac{87.5V}{(0,5T)(1m)} = 175\ m/s.$

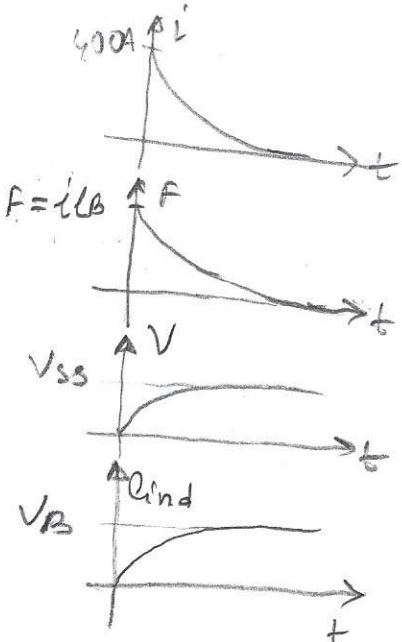
c) The efficiency

$P_{in} = V_B i = (100V)(50A)$

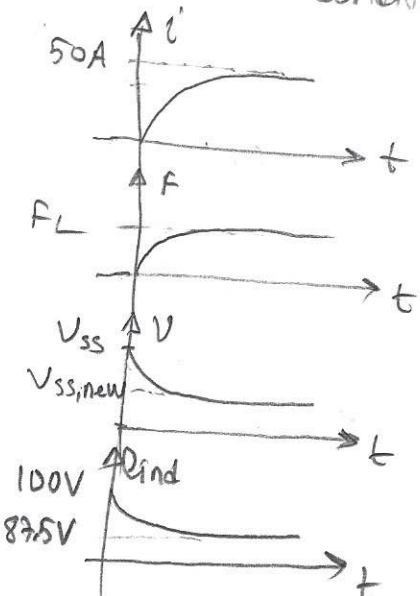
$P_{out} = F v = (25N)(175)$

$\eta = \frac{P_{out}}{P_{in}}(100) = 87.5\%$

d) At No load (b)



Under load condition (c)



3) 100-hp, 250-V, 1200 r/mm SHUNT DC MOTOR

$R_A = 0{,}03\,\Omega$, $R_F = 41.67\,\Omega$

No armature reaction, mechanical and Core losses negligible

$I_L = 126\,A$, $n_{initial} = 1103$ r/min

$I_A \rightarrow$ constant

n ? if R_F is raised to $50\,\Omega = R_{F2}$

① $I_{A1} = I_{L1} - I_{F1} = I_{L1} - \frac{V_T}{R_{F1}} = 126V - \underbrace{\frac{250V}{41.67\Omega}}_{6A} = 120\,A$

$I_{F1} = 6A \rightarrow E_{A1,MC} \cong 268V$ (from the magnetization curve)

② $I_{A1} \Rightarrow E_{A1,real} = V_T - I_{A1}R_A = 250V - (120A)(0{,}03\Omega) = 246.4V$ (real value under present condition)

③ $I_{F2} = \frac{V_T}{R_{F2}} = \frac{250V}{50\Omega} = 5A \rightarrow E_{A2} = 250V$ (from the curve)

④ Since I_A is constant, E_A remains constant, looking at ②

Hence $\frac{E_{A2}}{E_{A1}} = \frac{K\phi_2 n_{m2}}{K'\phi_1 n_{m1}} = 1 \Rightarrow n_{m2} = \frac{\phi_1}{\phi_2} n_{m1}$

(E_{A1} present condition)

⑤ $E_{A1} = E_{A2}$

$E_{A2,MC} = 250V$ at $I_{F2} = 5A$ and $E_{A1,MC} = 268$ at $I_{F1} = 6A$ (From the Fig 1) are directly proportional to the flux, now the ratio of $\frac{\phi_1}{\phi_2}$ will approximately equal to the ratio of $\frac{E_{A1,MC}}{E_{A2,MC}}$



$$\frac{\phi_1}{\phi_2} = \frac{E_{A1,MC}}{E_{A2,MC}} = \frac{268V}{250V} = 1.072$$, and from ④

$n_{m2} = (1.072)(1103\,r/min) = 1182.4\,r/min$

4) Sepately excited DC G. ∋ 172kW, 430V, 400A, 1800 r/min.

$R_A = 0{,}05\,\Omega$, $R_F = 20\,\Omega$, $R_{adj} = 0$ to $300\,\Omega$, $V_F = 430V$, $N_F = 1000$ turns/pole.

a) $R_{adj} = 63\,\Omega$ $n_{prime} = 1600$ r/min $V_{T,no\,load} = ?$

b) V_T if $I_L = 360A$, with compensating windings

c) V_T if $I_L = 360A$, without " "

$F_{AR} = 450$ A.turns

d) $I_{F,new} = ?$ to restore V_T to the original value (no load voltage) with existing compensating winding

$R_{adj} = ?$ (the required value of R_{adj})

a) $V_{T,no\,load} = E_{A,1600} \Rightarrow$ $I_f = \frac{V_F}{R_F + R_{adj}} = \frac{430V}{20\Omega + 63\Omega} = 5.18A$

$\frac{E_{A,1600}}{E_{A,1800}} = \frac{n_m}{n_o} \Rightarrow E_{A,1600} = \frac{n_m}{n_o} E_{A,1800} = \frac{1600}{1800}(430V)$

$= 382V$ (V_T no load)

$I_F = 5.18 \Rightarrow E_{A,1800} = 430V$ (from the MC (Fig 2)

$V_{T,NL} = 382V$

b) $V_T = E_A - I_A R_A = 382V - (360A)(0{,}05\Omega) = 364V$

c) $I_F^* = I_F - \frac{F_{AR}}{N_F} = 5.18 - \frac{450\text{ A.turns}}{1000\text{ turns/pole}} = 4{,}73A$

from the MC $E_{A0} = 410V$ at 1800 r/min

$E_{A,1600} = \frac{n_m}{n_o} E_{A0} = \frac{1600\text{ r/min}}{1800\text{ r/min}}(410V) = 364.4V$

$V_T = E_{A,1600} - I_A R_A = 364.4V - (360A)(0{,}05\Omega) = 346.4V$

4-) d) $I_F = ?$ to raise V_T to the no load value

$$E_{A,1600} = V_T + I_A R_A = 382V + (360A)(0,05\Omega) = 400V$$

$$\frac{E_{A,1600}}{E_{A,1800}} = \frac{n_m}{n_o} \Rightarrow E_{A,1800} = \frac{n_o}{n_m} E_{A,1600} = \frac{1800}{1600}(400V)$$

$$E_{A,1800} = 450V.$$

From the MC the corresponding $I_F = 6.15A$

$$I_F = 6.15 = \frac{V_{T,NL}}{R_F + R_{adj}} = \frac{430V}{20 + R_{adj}} \Rightarrow R_{adj} = \frac{430 - (6.15)(20)}{6.15}$$

$$= 49.9 \approx 50\Omega.$$